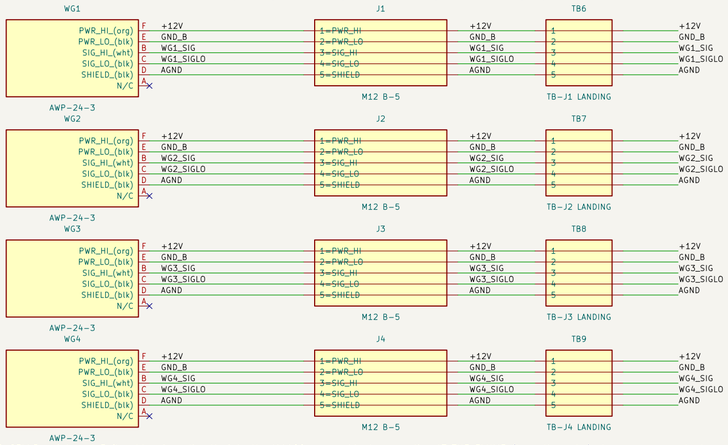
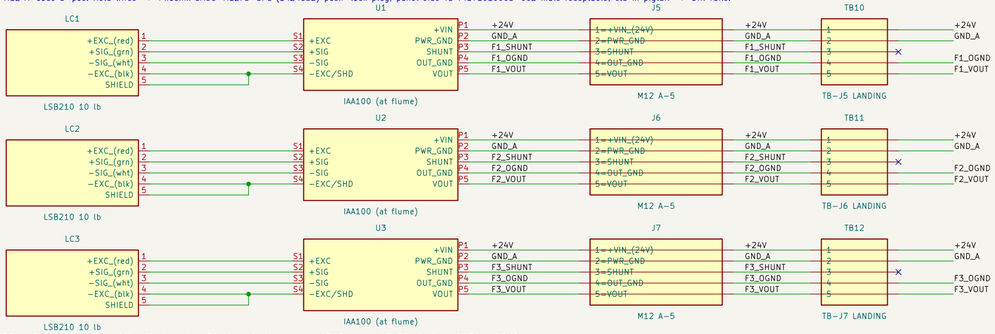
4x WAVE HEIGHT GAUGES — runs W1–W4 = twisted bundles of 5 single 20 AWG wires (twist SIG_HI wht with SIG_LO blk). Gauge bonds PWR_LO to SHIELD internally.

J1–J4 = M12 B-code 5-pos: Phoenix SACC-M12FSB-5PL SH (1424664) push-lock plug on the wire bundle (heat-shrink at the gland): TE T4171010405-002 receptacle -> DIN rails.



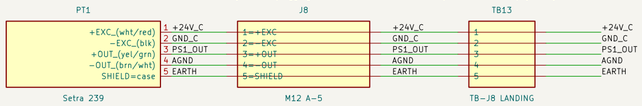
3x LOAD CELLS — integral 5 ft cable wires directly into the IAA100 sensor block: amps live AT THE FLUME (keep them dry). Amp power + output pass through J5–J7.

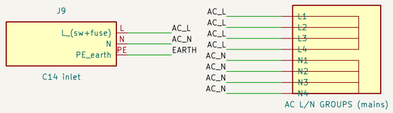
J5–J8 = M12 A-code 5-pos: field wires -> Phoenix SACC-M12FS-5PL (1424652) push-lock plug: panel side TE T4171010005-002 male receptacle, 0.5 m pigtail -> DIN rails.



PRESSURE — Setra 239 plugs into J8. Isolated 24 V excitation only: −EXC must NEVER tie to −OUT or AGND.

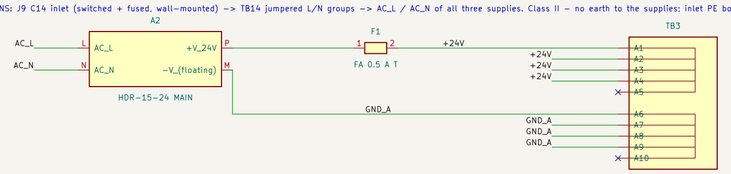
J8 uses the SAME connector as J5–J7 — label all four clearly; cross-plugging misroutes the 24 V domains. (B-coded J1–J4 physically cannot mate these.)



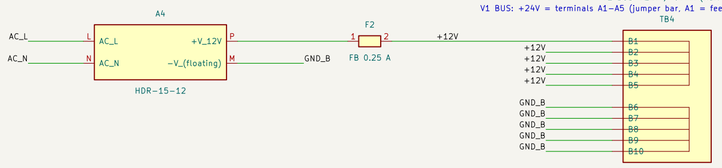


POWER (Rail A) — three Class II supplies, all outputs floating: A2 HDR-15-24 -> FA (time-delay) -> V1 bus (GND_A); A4 HDR-15-12 -> FB -> V2 bus (+12 V, GND_B); A3 HDR-15-24 = the Setra's private island -> X1–X2 (GND_C).

MAINS: J9 C14 inlet (switched + fused, wall-mounted) -> TB14 jumpered L/N groups -> AC_L / AC_N of all three supplies. Class II — no earth to the supplies; inlet PE bonds to the panel earth stud.



V1 BUS: +24V = terminals A1–A5 (jumper bar, A1 = feed from FA, A2–A4 = the 3 amps, A5 spare) / 0V-A = A6–A10 (A6 = feed, A7–A9 = amps, A10 spare).



V2 BUS: +12V = terminals B1–B5 (jumper bar, B1 = feed from FB, B2–B5 = the 4 gauges) / 0V-B = B6–B10 (B6 = feed, B7–B10 = gauges).

Three power domains, three returns: GND_A (force — grounded via the amps' internals), GND_B (gauges — grounded via their shield bonds), GND_C (pressure — referenced only by Setra −Out on the GND bus).

TB-J1..TB-J8 landing strips (Rail C): pigtail in, internal wire out; terminal numbers = M12 contacts.

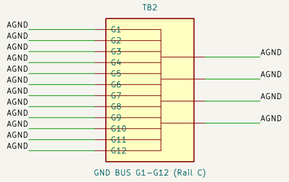
J1–J4 'shield' = a black wire from gauge pin D: land it on contact 5, NOT on the SH shield shell — the single-point ground lives at the GND bus only.

M12 pigtail color code, A- and B-code alike (verify on the TE insert): contact 1 = brown, 2 = white, 3 = blue, 4 = black, 5 = gray. Label each lead before trimming.

Assembly path: sensor -> (amp at flume) -> bulkhead J -> TB-J landing strip -> internal panel wire -> bus / DAQ. Terminal IDs match Interface Doc sec.4.

EARTH = panel earth stud (mains inlet PE + Setra case/shield). SHUNT leads park on TB-J5..J7 terminal 3 (back side empty): touch to +24V momentarily for shunt cal.

GND BUS G1–G12 (Rail C, all jumpered) — shields, Setra −Out and the GND_A reference jumper collect here: 2 drops out to DAQ AGND pins 12 and 39.



DAQ HOOKUP (DIFFERENTIAL) — every channel = a twisted WHITE+GREEN pair from its landing strip. WHITE = signal HI -> CHxH, GREEN = signal LO -> CHxL. Wire by terminal NUMBER (silk), not by color.

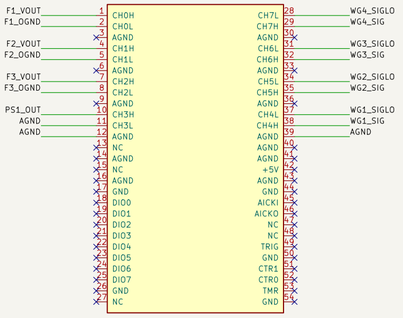
1. TB-J5: term 5 (Vout) -> pin 1 CH0H, term 4 (OutGnd) -> pin 2 CH0L.
2. TB-J6: term 5 -> pin 4 CH1H, term 4 -> pin 5 CH1L.
3. TB-J7: term 5 -> pin 7 CH2H, term 4 -> pin 8 CH2L.
4. TB-J8: term 3 (+Out) -> pin 10 CH3H, term 4 (−Out) -> pin 11 CH3L. Term 4 ALSO keeps its GND-bus wire (isolation island reference).
5. TB-J1..J4: term 3 (SigHi) -> CH4H–CH7H = pins 38, 35, 32, 29. Term 4 (SigLo) -> CH4L–CH7L = pins 37, 34, 31, 28. REMOVE the old SigLo wires to the GND bus.
6. Two GREEN jumpers from the GND bus to pin 12 AGND and pin 39 AGND. 7. NEW reference jumper: one BLUE wire A6 (GND_A) -> GND bus.

Shields (TB-J1..J4 term 5) stay on the GND bus. Each power island touches AGND exactly ONCE: gauges via shields, force via the A6 jumper, Setra via −Out.

SOFTWARE: InstaCal -> device -> Analog Input Mode = DIFFERENTIAL. DAQami ranges: CH0–CH2 +/−10 V, CH3 +/−5 V (Setra is +/−2.5 V), CH4–CH7 +/−10 V. 15 min warm-up before

Every terminal marked X stays EMPTY (unused AGND / DIO / counter / clock pins; 13, 15, 27, 47, 48 are NC). NEVER land a supply wire on the DAQ: max input is +/−25 V to AGND.

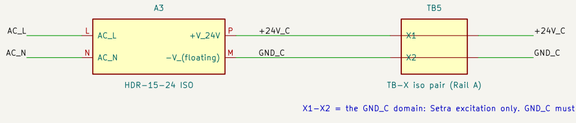
USB-1608G — DIFFERENTIAL mode (16-bit, range software-selectable per channel). USB to lab PC.



AGND = the system's only measurement ground. GND bus drops to DAQ pins 12 and 39.

GND_A ——— AGND

REFERENCE JUMPER (new for differential); one BLUE wire from A6 (GND_A) to the GND bus.
The amps' OutGnd pins are now CH0L–CH2L inputs, so this is the force island's only path to AGND.



X1–X2 = the GND_C domain: Setra excitation only. GND_C must NEVER be jumpered to GND_A, GND_B, or the GND bus.

gauges via J1–J4 (M12 B-code, cannot cross-mate)
DIN-rail strips shown; amps at flume via J5–J7 (M12 A-code);

Sheet: /		
File: wave_flume_daq.kicad_sch		
Title: Wave Flume DAQ — Sensor Interconnect		
Size: A2	Date: 2026-09-10	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1